

Block diagrams and analysis of feedback

R0005E - Measurement and Feedback Control

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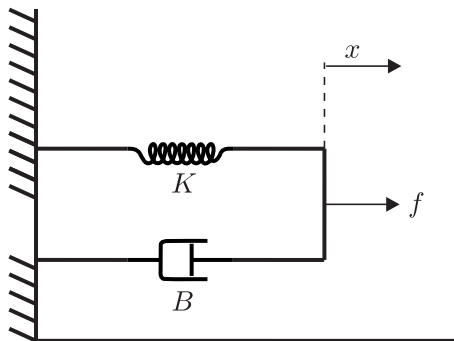
System modeling diagrams (Visualization)

Differential equations are not always intuitive to understand or to discuss with others. The visual representation is therefore important to improve understanding.

There are different types of modeling diagrams:

- **Block diagrams:** The elements are described as blocks and the arcs are the signals that connect the elements. The signals can be more or less anything, there is no convention determined. This is also the representation that is used in Matlab/Simulink.
- **Bond graphs:** Describes a system in terms of the energy that is transmitted between the elements. The arcs are bidirectional and indicate the transmission of energy. There is no need to determine the flow direction.
- **Signals flow graphs:** Here the elements are described by the arcs and the signals are the nodes in the graph. The block diagram and the signal flow graph are very related and be directly transformed into each other.

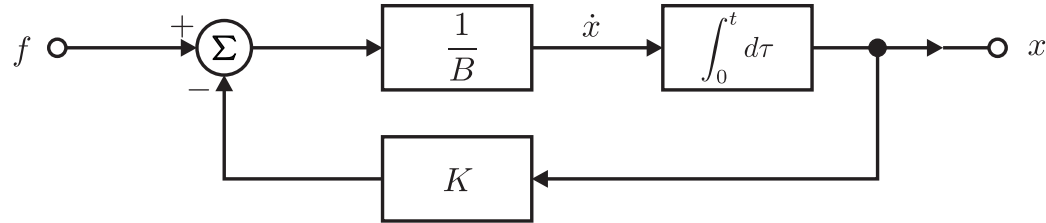
Remember the spring-damper example:



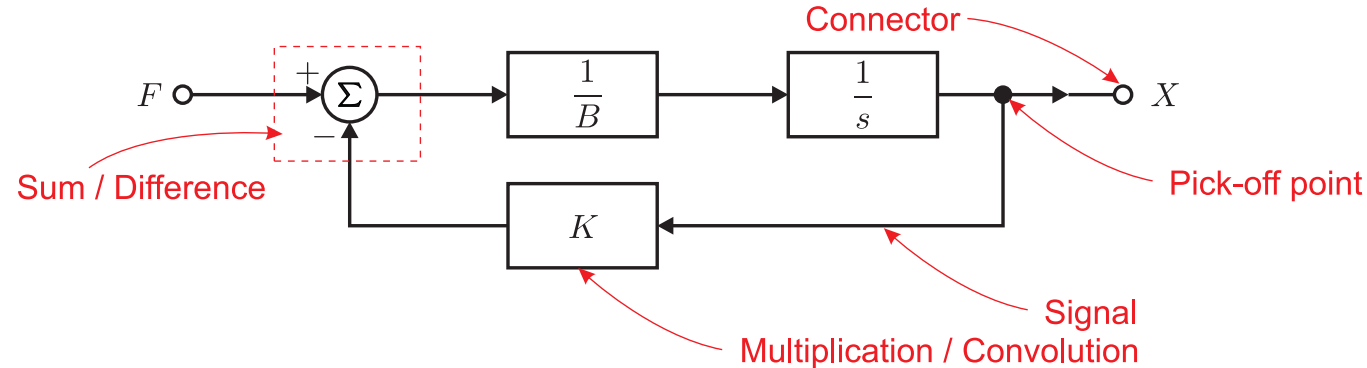
A bond graph representation of the spring-damper system. It consists of a vertical line with two outgoing bonds to the left. The top bond is labeled Kx and the bottom bond is labeled Bv . An incoming bond from the right is labeled f . The diagram is followed by an implication arrow $\Rightarrow$ and the differential equation.

$$\dot{x}(t) = -\frac{K}{B}x(t) + \frac{1}{B}f(t)$$

The system can also be represented graphically as the following block diagram:



In the same way as the differential equation can be transformed to the Laplace domain, the block diagram can be transformed and yields:



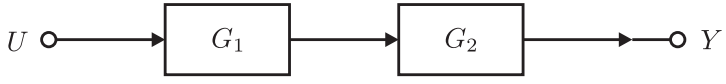
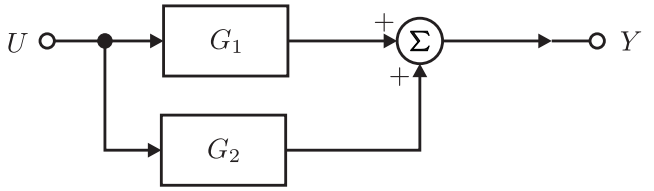
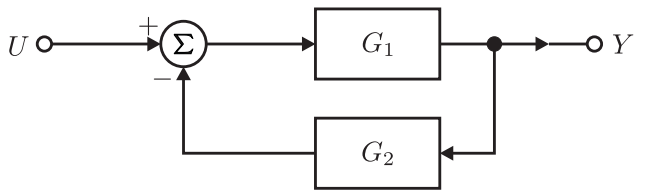
Here the notation for the different elements in the block diagram are also given in red.

Note: The **Connector** is not very often used. It can be omitted.

Block diagram algebra

Even for block diagrams there exist algebraic operations. Those can always be related and derived from algebraic operations on equations.

For the sake of simplicity, the operator s is dropped in both transfer functions and variables.

Block diagram operation	Algebraic operation
	$Y = G_2 \cdot G_1 \cdot U$
	$Y = (G_1 + G_2) \cdot U$
	$Y = \frac{G_1}{1 + G_2 G_1} U$

Notes:

- The blocks in the left column can be replaced by one block!
- The last operation is of large importance when it comes to feedback systems.

The last operations could be either used to compose large blocks or to decompose blocks into several minor blocks. Additionally, a block diagram can be restructured or manipulated without affecting its input-output characteristics:

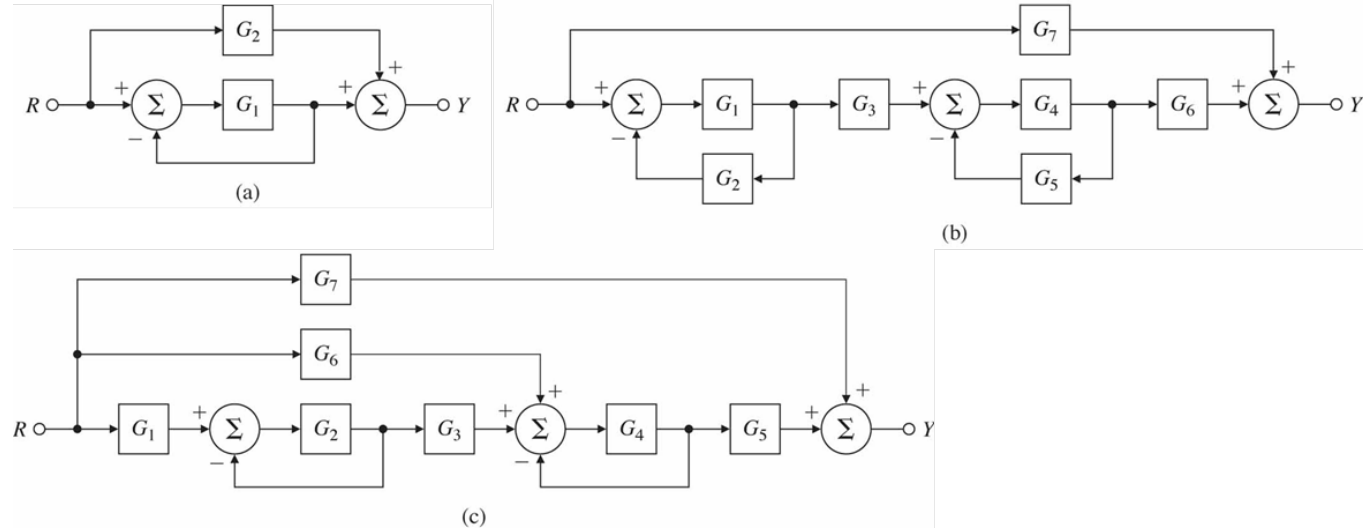
Manipulation	Block diagrams
Moving a pick-off	
Moving a sum	
Moving a feed-back gain	

Notes:

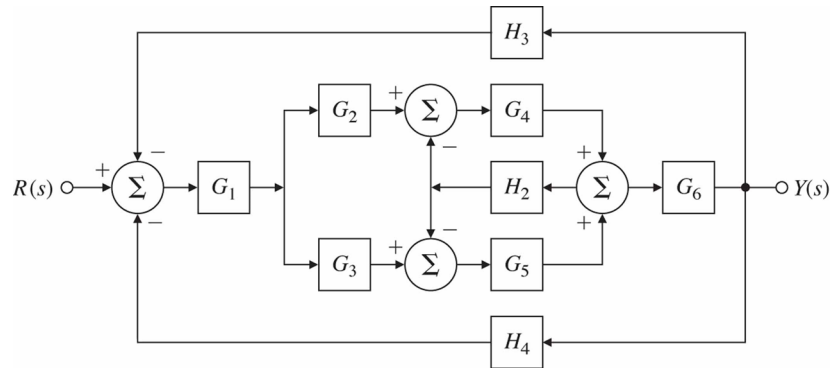
- Manipulation of a block diagram can also be done by resolving the equations directly.
- In order to derive the equation governing the block diagram, add an auxiliary variable after each sum. Then derive equations for all variables and solve the equation system.

Recommended problems

You can practice the derivation of the transfer function from a block diagram in problems 3.20:



and problem 3.22:



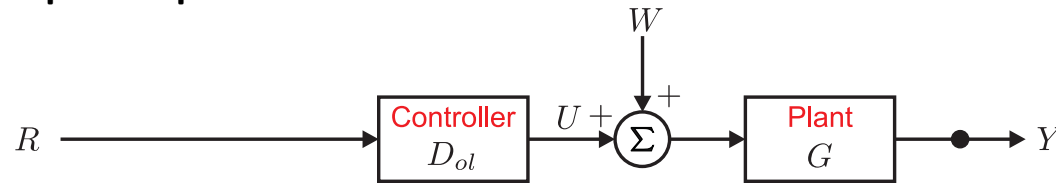
Analysis of feedback systems

Until now we have been mainly focusing on the system alone and disregarding that there might be a controller present.

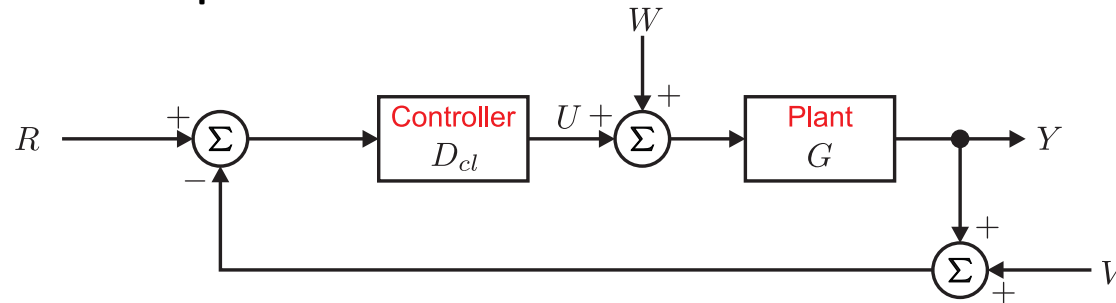
From the block diagram manipulation we have seen that a block diagram can represent a complete control system, comprising controller and system.

Principally, there are two kinds of control systems:

1. Open-loop:



2. Closed-loop:



No matter how the control system is structured, the aim of the control system is usually defined in terms of certain properties or abilities.

Properties/Abilities:

- Tracking: Ability of the output Y to follow a reference input R .
- Regulation: Ability to keep the control error small despite step like references and presence of disturbances.
- Stability: Property of the control system.
- Sensitivity: Ability to keep the control error small despite the presence of model uncertainties.

Control error:

The control error is defined as the deviation of the achieved output signal to the desired output signal. The desired output signal is usually given in terms of a reference signal:

$$E = R - Y$$

Observation:

The control error E is a signal in the closed loop system but not in the open loop. Question: How does this affect the control system properties and abilities?

Open loop vs. Closed loop

We have a look at the following simple example to compare the open loop with the closed loop.

Room temperature control:

Assume that a room has a heater where the effect P can be adjusted and a thermometer that can measure the temperature T in the room. The walls of the room will leak heat P_O to the outside, depending on the outside temperature T_O . It is assumed that the room has an effective heat capacity C and the walls have a thermal resistance R_{th} .

Task: Keep the room temperature at a desired constant level.

First we derive a dynamic model of the room:

- Change in the air heat content: $\dot{Q} = C\dot{T}$
- Heat leakage of the wall: $P_O = R_{th}(T - T_O)$
- Heat balance: $\dot{Q} = P - P_O$

We can combine the equations and get a complete model

$$C\dot{T} = P - R_{th}(T - T_O) \quad (1)$$

If we now assume that the room has a very small heat capacity ($C \approx 0$), then (1) simplifies to

$$T = \frac{1}{R_{th}}P + T_O$$

Now we choose the following controllers:

1. Open loop case: $D_{ol} = R_{th}$
2. Closed loop case: $D_{cl} = K$, which is a constant gain.

We will now analyze the tracking behavior of both control systems, which means that we analyze the behavior of the control error E .

1. Open loop case:

$$\begin{aligned} E_{ol} &= T_{ref} - T \\ &= T_{ref} - \left(\frac{1}{R_{th}} R_{th} T_{ref} + T_O \right) \\ &= -T_O \end{aligned}$$

2. Closed loop case:

$$\begin{aligned} E_{cl} &= T_{ref} - T \\ &= T_{ref} - \left(\frac{1}{R_{th}} K E_{cl} + T_O \right) \\ \Rightarrow E_{cl} &= \frac{1}{1 + R_{th} K} T_{ref} - \frac{1}{1 + R_{th} K} T_O \end{aligned}$$

Analysis:

Obviously, in the open loop case the control error does directly depend on the outside temperature T_O and the controller has no effect on this. Still, the controller will achieve perfect tracking since the controller does not depend on T_{ref} .

In the closed loop case, the control error does depend on the controller gain K . For a large value of K , the control error will be small both for tracking and regulation.

Number example:

We use the following settings:

- $T_{ref} = 5$ (this means a raise of 5 degrees around the working point)
- $R = 1$
- $K = 100$

Now we assess the control systems for the two disturbance cases:

1. The outside temperature is at the working point level: $T_O = 0$.
2. The outside temperature is dropping by $T_O = -10$.

Result:

1. $T_O = 0$

- Open loop: $E_{ol} = 0$
- Closed loop: $E_{cl} = \frac{5}{101} \approx 0.05$

2. $T_O = -10$

- Open loop: $E_{ol} = -10$
- Closed loop: $E_{cl} = \frac{5}{101} + \frac{10}{101} \approx 0.15$

Conclusions:

- Closed loop control achieves both good disturbance rejection and good tracking abilities.
- Open loop does yield perfect tracking but has very poor disturbance rejection ability.
- Here the controller was just chosen in an ad-hoc manner. We need a more systematic way to find the controller.
- Specifications on the control error could be used within the controller design.

Generalisation of the control system equations

In order to understand the differences between the open and closed loop case in general, we can derive transfer functions that describe the dynamic behavior between different variables in the control system.

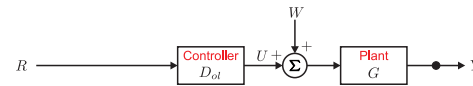
We are interested of the behavior of the control error E , the output signal Y , but also of the control action U . We will now derive transfer functions for these signals.

1. Open loop case:

$$Y_{ol} = GD_{ol}R + GW \quad (2)$$

$$\begin{aligned} E_{ol} &= R - Y_{ol} \\ &= R - (GD_{ol}R + GW) \\ &= (1 - GD_{ol})R + GW \end{aligned} \quad (3)$$

$$U_{ol} = D_{ol}R \quad (4)$$

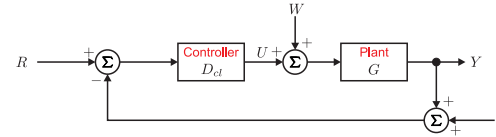


Obviously, choosing $D_{ol} = G^{-1}$, would yield perfect tracking. But, this implies that the controller would create infinite control actions for plants G which do not instantaneously react on control actions.

Thus, this is not feasible!

2. Closed loop case:

$$Y_{cl} = \frac{GD_{cl}}{1 + GD_{cl}}R + \frac{G}{1 + GD_{cl}}W - \frac{GD_{cl}}{1 + GD_{cl}}V \quad (5)$$



$$\begin{aligned} E_{cl} &= R - (Y_{cl} + V) \\ &= R - \left(\frac{GD_{cl}}{1 + GD_{cl}}R + \frac{G}{1 + GD_{cl}}W - \frac{GD_{cl}}{1 + GD_{cl}}V + V \right) \\ &= \frac{1}{1 + GD_{cl}}R - \frac{G}{1 + GD_{cl}}W - \frac{1}{1 + GD_{cl}}V \end{aligned} \quad (6)$$

$$U_{cl} = \frac{D_{cl}}{1 + GD_{cl}}R - \frac{GD_{cl}}{1 + GD_{cl}}W - \frac{D_{cl}}{1 + GD_{cl}}V \quad (7)$$

Some conclusions:

- A controller with a high gain will yield good tracking, but will yield large control actions, which might not be possible to achieve.
- In order to be able to implement ("build") a controller it has to be **proper**, which means that the order of the denominator of needs to be higher than or equal to the order of the numerator.
- Disturbance rejection ability is the general advantage of the closed loop setup.

Stability

Remember: In order for a transfer function to be stable, all the roots of the characteristic polynomial need to reside in the LHP.

1. Open loop case:

Here there are two transfer function, (1) from R to Y and (2) from W to Y . In order for the control system to be stable G and GD_{ol} need to be stable.

Thus, if GD_{ol} has all its poles in the LHP, then the control system is stable.

2. Closed loop case:

It can be observed that the transfer functions (5), (6) and (7) have the same denominator:

$$1 + GD_{cl}$$

Thus, in order to have a stable closed loop system, the roots of

$$1 + GD_{cl} = 0$$

have to reside in the LHP.

Tracking

Remember: The tracking ability is reflected by the transfer function from R to Y .

1. Open loop case:

It can be seen that the transfer function for tracking the open loop case is

$$T_{ol} = GD_{ol}$$

Perfect tracking, is achieved if $GD_{ol} = 1$. But, we need to consider two important limitations:

1. Limitations in the control action
2. Properness of the controller

2. Closed loop case:

Similarly, we can define the transfer function for tracking in the closed loop case

$$T_{cl} = \frac{GD_{cl}}{1 + GD_{cl}}$$

Again, perfect tracking would require $T_{cl} = 1$, which is not possible to achieve in general, as $GD_{cl} \neq 1 + GD_{cl}$.

The same limitations as in the open loop case for the controller apply even here.

Regulation

Remember: The regulation ability target the rejection of disturbances W and noise V , as they are reflected in the control error.

1. Open loop case:

Open loop control does not have any regulation ability. Noise is usually not affecting the control scheme.

2. Closed loop case:

Since the output is fed back, the controller has the ability to react on disturbances and noise. In (6) the terms relating to noise and disturbance are:

$$-\frac{G}{1 + GD_{cl}}W, \quad \frac{GD_{cl}}{1 + GD_{cl}}V$$

Clearly, a controller that has good disturbance rejection ability should have a high gain (D_{cl} large). But at the same time this means that the noise is not rejected.

This is a typical trade-off that can be seen in all closed loop control systems. The controller gain need to be limited.

Sensitivity

Remember: Ability to keep the control error small despite model uncertainties.

Model uncertainties could be described by a perturbation of the plant G with δG . Thus, the real plant would be $G + \delta G$, and the relative perturbation is $\delta G/G$.

The sensitivity can now be defined in relative terms

$$S_G = \frac{\frac{\delta T}{T}}{\frac{\delta G}{G}}$$

1. Open loop case:

Here, $T_{ol} = GD_{ol}$ and consequently, $\delta T_{ol} = \delta GD_{ol}$, which results in

$$S_G = \frac{\frac{\delta T}{T}}{\frac{\delta G}{G}} = 1$$

Thus, the sensitivity of the open loop system to model uncertainties is 100%. Any uncertainty is fully reflected in the control error.

2. Closed loop case:

In contrary to the open loop case, any perturbation which is reflected in the output will cause the controller to react and will therefore lead to an attenuation.

It can be shown that

$$S_G = \frac{\frac{\delta T}{T}}{\frac{\delta G}{G}} = \frac{1}{1 + GD_{cl}}$$

This means that the controller design has an effect on the way the closed loop system will react on model uncertainties and how it will attenuate its effects.

Notes:

- S_G is the transfer function from R to E !
- Reduced sensitivity to model uncertainties is one of the major advantages of feedback!

Steady state properties

Although, the abilities like tracking and regulation are now defined, there is no direct quantification given.

Principally, the solutions for the equations (5), (6) and (7) can be derived and analyzed in the time domain.

Instead, the steady state information can be found in the Laplace domain using the final value theorem. We define:

- Steady state control error:

$$e_{ss} = \lim_{s \rightarrow 0} sE(s)$$

- Steady state output value:

$$y_{ss} = \lim_{s \rightarrow 0} sY(s)$$

The transfer function for both tracking and regulation enable us to analyze the dynamic and steady state behavior of a control system.

Notes:

- System type is not discussed in this course. Since e_{ss} and y_{ss} are straight forward to derive, it is not a necessity for the control design and control system analysis.
- The filtered case is not further discussed. All transfer functions can be adapted for the filtered case, straight away.

Additional material

The following videos are recommended in relation to this lecture, but also relating to lecture 6.

- Introduction to System Stability and Control
<https://www.youtube.com/watch?v=uqjKG32AkC4>
- Stability of Closed Loop Control Systems
<https://www.youtube.com/watch?v=yf090rHa520>
- Final Value Theorem and Steady State Error
<https://www.youtube.com/watch?v=PXxveGoNRUw>
I have referred to this video before, but it discussed system types and the steady state control error, so it could be good to look at it again.